OLIST E-COMMERCE PERFORMANCE ANALYSIS

Project Brief and Recommendations

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Tools	Microsoft Excel | MySQL | Power BI
Dataset	Brazilian E-Commerce Public Dataset by Olist
Deliverable	Three-page interactive dashboard with a dedicated tooltip page

1. Project Overview

This project analyses Olist's Brazilian e-commerce dataset to evaluate marketplace performance, product-category revenue, customer purchasing behaviour, delivery efficiency, payment patterns, and customer satisfaction. The analysis was presented in Power BI through an interactive report designed to help users explore performance and identify practical improvement opportunities.

2. Business Problem and Aim

Olist needs to understand how its marketplace is performing, which product categories and customer locations contribute most to revenue, how efficiently orders are processed and delivered, and how delivery experiences relate to customer reviews and repeat purchasing.

Project aim: To use Excel, MySQL, and Power BI to prepare and analyse e-commerce data, measure key business indicators, identify patterns in sales and delivery performance, and communicate findings through an interactive dashboard.

3. Data and SQL Workflow

The project used nine related CSV datasets. I used MySQL's LOAD DATA INFILE approach to import the CSV files into their corresponding database tables, making bulk loading faster than importing records manually. I then used SQL to query and analyse the data, and Power BI to build the dashboard.

Dataset / table	Purpose
olist_customers_dataset	Customer records and customer locations
olist_geolocation_dataset	Geolocation information
olist_order_items_dataset	Products, sellers, item prices, and freight recorded for order items
olist_order_payments_dataset	Payment methods, payment values, and instalments
olist_order_reviews_dataset	Customer review scores and review information
olist_orders_dataset	Order status and purchase, approval, carrier, delivery, and estimated-delivery timestamps
olist_products_dataset	Product details and category information
olist_sellers_dataset	Seller records and locations
product_category_name_translation	English translations of product category names

In Power BI, I created a separate date dimension table (dim_date) to support date-based filtering and monthly/yearly analysis. The source timestamps were used to calculate elapsed time between order stages, including approval time, processing time, delivery time, and overall fulfilment time.

4. Dashboard Structure

Page 1 — Olist E-commerce Analysis: Overall KPIs, order status, revenue by month, top and bottom product categories by revenue, customer-state and customer-city revenue, and payment types.

Page 2 — Delivery & Customer Experience: Approval, processing, delivery and fulfilment time; on-time delivery; delivery variance; review scores; delivery trends; and delivery performance by customer state.

Page 3 — Customer Retention & Delivery Insights: Repeat customers by product category, delivery status among repeat-customer orders, review scores by delivery status and payment type, and late deliveries by customer city.

Tooltip page — Order & Review Tooltip: A custom tooltip displaying total orders, average review score, late deliveries, and on-time deliveries when users hover over selected visuals.

Navigation buttons connect the three main report pages. Customer State and Year slicers let users interact with the report. The tooltip page supplies extra detail without overcrowding the main visuals.

5. Main KPIs and Definitions

KPI	Meaning
Total Customers	Distinct customers counted using customer_unique_id.
Total Orders	Distinct order IDs.
Total Products / Sellers	Distinct products in the product dataset and distinct sellers in the seller dataset.
Total Items Sold	Number of order-item records/units represented by the measure; different from the number of orders.
Total Revenue	Sum of product prices, excluding freight; it is not profit.
Total Freight	Sum of freight charges recorded on order items; it is not necessarily Olist's actual shipping expense.
Total Order Value	Product prices plus freight charges, based on the measures used in this report.
Average Order Value	Total Order Value divided by distinct orders when freight is included.
Average Approval Time	Average time from purchase to order approval, in hours.
Average Processing Time	Average time from approval to carrier handover, in hours.
Average Delivery Time	Average time from carrier handover to customer delivery, in days.
Average Fulfilment Time	Average time from purchase to customer delivery, in days; it includes the full order journey.
On-Time Delivery Rate	Percentage of classified orders delivered on or before the estimated date.
Average Delivery Variance	Actual delivery date minus estimated delivery date. A negative result means orders arrived earlier than estimated.
Repeat Customer Rate	Percentage of unique customers who placed more than one order under the project's definition.
Average Review Score	Mean customer review score on the dataset's 1–5 scale.

6. Key Findings

- The dashboard shows approximately 99,000 orders, 96,000 customers, 33,000 products, 3,000 sellers, and 113,000 items sold.
- Total product revenue is approximately R$13.59 million, total freight is R$2.25 million, and total order value is approximately R$15.84 million.
- Health and beauty, watches and gifts, and bed/bath/table are among the leading product categories by revenue.
- The on-time delivery rate is approximately 92%, with around 89,000 on-time orders and 8,000 late orders. Around 3,000 orders have unknown delivery status in the overall dashboard.
- Average delivery time is approximately 9 days, while average fulfilment time is approximately 13 days. Fulfilment includes the period before carrier handover as well as the delivery journey.
- Average delivery variance is approximately -12 days, meaning orders included in the calculation arrived around 12 days earlier than their estimated delivery dates on average.

- CDs, DVDs, and musicals has the highest average review score among the categories displayed (about 4.7), while security and services is much lower (about 2.5).
- Average review score is approximately 4.3 for on-time deliveries, 2.6 for late deliveries, and 1.8 for orders with unknown delivery status. This is an observed association, not proof that delivery status alone caused the difference.
- São Paulo has the highest late-delivery count among the cities displayed, followed by Rio de Janeiro.
- Bed/bath/table has the highest repeat-customer count in the category comparison (252), followed by sports/leisure (184) and health/beauty (151).
- The repeat-customer delivery visual shows 6,342 orders placed by repeat customers: 5,670 on time (89.4%), 429 late (6.76%), and 243 unknown (3.83%).
- Debit-card payments have the highest average review score in the payment comparison (about 4.2), but represent a much smaller volume than credit-card payment records. The not_defined category has only three records, so its average should be interpreted cautiously.
- Among customer states with more than 400 orders, states with on-time delivery rates below 88% generally show average review scores below 4.0 in the current table.

7. Recommendations

1. **Reduce late deliveries.** Late deliveries average about 2.6 in review scores, compared with about 4.3 for on-time deliveries. Improving delivery reliability should be a priority.
2. **Prioritise high-volume locations.** Focus delivery improvements on São Paulo and Rio de Janeiro, which have the highest late-delivery counts among the cities displayed.
3. **Improve the customer experience in low-rated categories.** Pay particular attention to security and services, which records an average review score of about 2.5.
4. **Build on categories with repeat purchasing.** Support continued engagement in bed/bath/table, sports/leisure, and health/beauty, which lead the repeat-customer category comparison.
5. **Target weaker delivery performance by state.** Prioritise states with on-time delivery rates below 88%, where the current table generally shows average review scores below 4.0.
6. **Maintain strong delivery performance.** Sustain the overall on-time delivery rate of about 92% and continue meeting or beating estimated delivery dates.

8. Project Scope

This project is primarily descriptive analytics, supported by exploratory comparisons. It summarises historical performance and highlights patterns across sales, product categories, delivery status, review scores, payment types, customer locations, and repeat purchasing. The findings describe observed patterns.

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